

Customer Churn Analysis for Telecom Industry

Introduction

This project aims to predict customer churn in a telecom company using machine learning. The goal is to help the company retain customers by identifying factors leading to churn.

Abstract

We used the IBM Telco Churn dataset and trained a Random Forest Classifier to predict churn. The model achieved good performance, and SHAP was used for feature importance explainability. We also segmented customers into At Risk, Loyal, and Dormant.

Tools Used

- Python (Pandas, NumPy, Sklearn, SHAP, Seaborn)
- Jupyter Notebook

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Steps Involved

1. Data Cleaning and Preprocessing
2. Exploratory Data Analysis
3. Feature Engineering
4. Model Training with Random Forest
5. Churn Prediction and Evaluation
6. Explainability using SHAP
7. Customer Segmentation and Recommendations

Conclusion

The model successfully identified churn-prone customers. Insights derived from SHAP values can help in designing targeted retention strategies. Future work can include real-time prediction and personalized offers based on churn probability.